Back

Module Quiz
Graded Assignment • 15 min

Due Sep 11, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the primary advantage of using non-blocking MPI communication? 1 / 1 point

- It prevents all deadlocks
- It allows overlapping of communication and computation
- It guarantees faster execution in all cases
- It eliminates the need for synchronization

Nice work

Correct, this is the main advantage of non-blocking communication.

2. What will be the output of the following MPI code snippet? 1 / 1 point

```
int rank, size;
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
MPI_Comm_size(MPI_COMM_WORLD, &size);
printf("I am process %d out of %d\n", rank, size);
```

- It will print the same message for all processes
- It will print different messages for each process, showing their rank and total size
- It will cause a compilation error
- It will only print a message for the root process

Nice work

Correct, this is what the code does.

3. What is the purpose of the 'tag' argument in MPI send and receive calls? 1 / 1 point

- To specify the data type being sent
- To indicate the size of the message
- To uniquely identify or label the message
- To determine the priority of the message

Nice work

Correct, tags help match specific sends and receives.

4. Which of the following is true about MPI_Recv? 1 / 1 point

- It is always non-blocking
- It always behaves synchronously
- It can be either blocking or non-blocking depending on message size
- It never blocks the program execution

Nice work

Correct, MPI_Recv always blocks until the message is received.

5. What is the main difference between global and point-to-point communication in MPI? 1 / 1 point

- Global communication is always faster
- Point-to-point communication involves all processes
- Global communication involves all processes, while point-to-point is between specific processes
- There is no significant difference

Nice work

Correct, this is the key difference.

6. In a simple performance model for MPI communication, what two main factors typically contribute to the total message time? 1 / 1 point

- CPU speed and memory bandwidth
- Network latency and transmission time
- Message size and number of processes
- Compiler optimization and hardware configuration

Nice work

Correct, these are typically the main factors considered.

7. What is a potential issue with using blocking MPI communication? 1 / 1 point

- It always results in race conditions
- It can lead to deadlocks if not implemented carefully
- It is always slower than non-blocking communication
- It requires more memory than non-blocking communication

Nice work

Correct, this is a common issue with blocking communication.

8. What is the purpose of MPI_Waitall in non-blocking communication? 1 / 1 point

- To initiate multiple non-blocking communications
- To check if any communication has been completed
- To wait for all initiated non-blocking communications to complete
- To cancel all pending non-blocking communications

Nice work

Correct, this is its primary purpose.

9. What will be the result of the following MPI code snippet? 1 / 1 point

```
int data;
MPI_Status status;
if (rank == 0) {
    data = 42;
    MPI_Send(&data, 1, MPI_INT, 1, 0, MPI_COMM_WORLD);
} else if (rank == 1) {
    MPI_Recv(&data, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, &status);
    printf("Received: %d\n", data);
}
```

- All processes will print "Received: 42"
- Only process with rank 1 will print "Received: 42"
- It will cause a deadlock
- It will cause a runtime error

Nice work

Correct, this is what the code does.

10. What is a key consideration when implementing parallel algorithms using MPI? 1 / 1 point

- Ensuring all processes have the entire dataset
- Using only collective communication methods
- Properly managing data distribution and inter-process dependencies
- Avoiding all forms of synchronization

Nice work

Correct, this is crucial for correctness and efficiency.